

JABchem

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Question Bank

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Chemistry

Self-Evaluation Question Bank

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Section 1a: Electromagnetic Radiation & Atomic Spectra

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
1 2 4															
3	mc1		mc1										(mc2) (mc3) (mc4)		(mc1) L1b(ii)
5															
6 7 9															
8 13 14				L5a	L1a L1b(ii)										
10 11 12	L1b L6a	L1b	L1a L5c(i)	L5c(i) L7d(ii)C	mc25 L1b(i)	L6c(ii)	L6a(iii)	L3b(ii)	L8a	L1c(iii)A L1c(iii)B			(L4d)	(mc6) L1a(ii)	L1b(i)
15 16														L1d	
17 18	L3b(ii)														
19 20		L1a(i) L1a(ii)	L5a	mc1	mc12	mc1 L1c	L1a(i)	L3b(i)	mc1	L1c(i) L1c(ii)				L1a(i)	(mc2)
21			L10c(i)A L10c(i)B	L3b(ii) L3b(ii) L3b(iii)						L2c			(mc7)	(mc7)	

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Section 1b: Atomic Orbitals and Electronic Configurations

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
22 23 24															
25			mc3												
26	L2b					L1a(i)									
27															
28a							L1a(ii)		mc2						
28b			mc2	mc4	mc2	L1a(ii)				mc1				mc1 L1c(iii)	
28c	L2c					L1a(ii)		L3a(ii)		L1b					
28d		mc3 L1b(ii)	L2b(ii)	L1a		L1a(ii) L5b(ii)A	L1a(ii)								L1a(iii)
29a	mc2		L1b(i)		mc1	L5b(i)								mc2 (mc10)	
29b														L1c(ii)	(L1a(ii))
29c								L3a(i)					mc5	mc5 L1b	
30													L1a		(L1a(i))
31															
32	L2a	mc2					mc1	mc1	**L9**	L1a			mc10 L1b	L1c(i)	
33	mc3						mc3								
34 35 36						L1b							(mc1) (mc6)		
37 38													(mc8)		
39 40					L11b	L3b(iii)		mc6		mc4					
41 42	mc4	L2b(i)	mc4	mc5	L2a		mc2	mc6	mc4				(mc9)	(mc8) (mc9)	(mc4) mc5
43		L2b(ii)						L8b							

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Section 1c: Transition Metals

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
44								L9a							
45		L4a(ii)	L10b	mc6	mc3		L5a	mc2		mc2				mc3	
46	L3a(iii)				L5a(i)	mc2			mc3				L5c(ii)	(mc11)	(mc3)
47 48		L4a(i)	mc6	mc7	L5a(ii)	L2c	L1a(iii)A	mc3	mc7						
49 50															
51 52						mc5		mc3					(mc10)		
53															
54 55		L6a(ii)	L6a L6b(ii)A L6b(ii)B	mc8	L2c(ii)	mc3	L5c(i) L8a(iii)	L3c(ii)		L5a(i)B L5a(i)C					mc7
57	L3b(i)	mc4 L6a(i)			L2b		L5c(ii)	L3c(i)	L10a(ii)A	mc3			L5a		mc8
56 58	mc5 mc16	mc5		L5b(i)	L2c(i)	L5b(ii)B	L5a L5b	mc5	L10a(ii)B L10b(i)	L3a(ii)A			L5c(i)		(mc6)
59 60 61															
62	L3a(ii)														
63 64 65	L3a(i) L3b(iii)		mc5	L5b(ii)		L5a L5b(ii)C	L5d(i) L5d(ii)	mc4	mc5 mc6 L6b(i)	L3a(ii)B			L5d(i) L5d(ii) L5d(iii)	mc4	mc9
66 67 68			L4a												
69a		L4b(ii)B					mc4								
69b 70						mc4	mc4	L10b							

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Section 1d: Chemical Equilibrium

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
71 72			mc7 mc10	L1c		mc7 L4a(ii)	L7a						mc13 (mc18)		
73			L2a(i)		L11a(i) L11a(ii)	L11b(i)B	L6a(i)	L8a(i)	L5d(ii)	L10b(ii)BI					
74 76		L2a(i)					L6a(ii)	L8a(ii)		L10b(ii)BII					(L6a(i))
75	L7a									mc8					
77															
78		L2a(ii)		mc9		mc7		L8b(iii)							
79								L8b(iv)		mc9					(L6a(ii))
80															
81															mc4
82															
83									L6b(ii)						
84 85						mc6									
86															
87															
88			mc9					mc8					L3		
89 90 91	mc6		mc8	L4a(i) L4a(ii) L4b	L8a		mc6 L7b(ii)		mc8	mc6 L7b(ii)B					
92						L3a									
93 94						L4d									
95		L6b(ii)	L2a(ii)							L10a(i)			mc17		
96															
97								mc9	mc11						
98	L5a			L4c(i)	mc4				mc10 L2b(i)						
99	L5b	mc7	L2b(i)	L4c(ii)B		L3b(i)	L7b(i) L7c(i)+(ii)	L7a(ii)	mc9 L2b(ii)	mc5 L10a(ii)			(L6a)		
100	L9b(iii)		mc11				mc5								mc20
101		L6b(i)													
102					L12f(i)										
103										L9❸					
104	mc8		mc12												(mc21)
105				L4d (3marks)	L8b(ii) L8c			mc7						(L13b)	
106					L8b(i)			mc10					(L6b)	(L13a)	
107										L9❸					
108															
109 110 111 112 113	mc7			L4d (3marks)					L2c(ii)						
114		mc6						L11a(ii) L11a(iii)		mc7			(mc18)		

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Section 1e: Reaction Feasibility

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
115	mc9				mc6	mc5			mc13						
116															
117	L1a(i)	L3a(i)A	L4d(i)A	mc11			L1b(i)A	L10a(i)	L4a(ii)B					(L3a(i))	(L2a)
118 119 120 121	mc10			mc12		mc11	mc9	mc12	mc14	mc11			(mc20)		
122					L1c(i)				L4a(ii)B				mc19		
123 124															
125															(mc22)
126										L3a(i)B					
127	L1a(ii)	L3a(i)B	L4d(i)B				L1b(i)B			L2a(i)			(L2a)	(L3a(ii))	(L2b)
128		mc8	mc13	L1b		L2a		L10a(ii)							
129				mc10			mc7	mc11		mc8 L3a(i)A					(mc23)
130															
131					mc5		mc8						(L2b)		
132 135															(mc24)
133	L1b	L3a(ii)	L4d(ii)		L1c(ii)		L1b(i)C	L10a(iii)	L4a(ii)C	L12a(ii)				(L3b)	(L2c)
134															
136															

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Section 1f: Kinetics

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
137															
138 139	L6b(i)	mc9		L2a	mc7 mc8	mc9	L2a(i) L2a(ii)		L12a(i) L12a(ii)	mc12 L6b(i)			(L8a)	(L11a)	(L6b(i))
140 141 142	L6b(ii)	L4b(i)	L9d(i)	mc13 L2b(ii)			L2b		L12a(iii)				(L8b)	(L11b(i))	
143				L2b(i)	L2d(i)			L1a(i)							
144						mc10		mc13						(mc13)	
145	L6b(iii)	mc10	L9d(ii)	mc14	L2d(ii)	L2d(i)B	L2c(i) L2c(ii)	L1a(ii)	L12a(iv)	L6b(ii)			(L8c)	(L11b(ii))	mc26 (L6b(ii))
146 147	mc11	L4b(ii)A	mc14	L2c	mc9	L2d(i)A L2d(ii)	mc10	L1a(iii)	mc12	mc13				(mc14)	
148															

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Section 2a: Molecular Orbitals

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
149 150															
151 152															mc10
153 154															
155 156 157															
158															
159				L7b(i)				L4b		L7a(ii)AII				(mc15)	
160	L9a(iii)							L4b							
161															
162b 163		mc11					mc11 L10a(ii)						L1c		
162a 164 165				L7b(ii)	L3a(ii)				L2a						
166			mc15			mc17									
167	mc12	mc12		mc15	mc12	mc13	L1a(iii)B		mc15 L6a(i)	mc14					
168 169						L6a(ii)									
170															
171 172							mc12		L2c(i)						
173 174 175			L5a	L7c		L6a(i)	L10a(i)	L4a(i)							
176			L5b L5c		L3a(iii)	L6c(i) L7b(iii)	mc12	L4a(ii) L4c	mc24	mc15					mc18

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Section 2b: Synthesis Page 1

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
177 178															
179 182a	L8c				mc11			L12c(i)	L8b(i)				L10b		
180 182b															mc13
181 183									L8b(ii)						
184 185 186		mc14			mc10	L8b(i)				L7a(ii)AI					L11a(i)
187 188 189					L12e										
190	mc18						L8b(ii) L10a(iii) L11b	L10c		L11b(ii)A			(mc28)		L4a
191															
192 193	mc13 L7b(iii) L9a(i)	L8a(i)	mc16 L6b(i)	mc16 L10b(i)	L12a	mc17	L6c L8a(i)+(ii) L8b(i)	L7a(i) L10d L12a	L7a	mc21 L10b(i)			L7a L7b		mc11 (L9a+d) (L9d(ii))
194 196					mc19	mc18		mc14 mc19	L11b(ii)B Part II				mc30		(L10a) (L10d)
195 197										L7c(i)					
198 199									mc17					mc18	
200 222			L9a		L12d	L9				mc19			mc23 (L10d)	(mc19)	(mc14) (L10b)
201a			L8d		L7d(i)	L9	mc14			mc19			(L8e) (L10c)		
201b 211	mc16		mc18	L10c(i)	L7a					L6a(i) L6a(ii)			mc23	(mc21)	mc15
201c	mc20	L9b	mc19	mc22	L7b L7c	mc16								(L9b(iii))	
202 206	L8d				L9d			L12c(ii)		L7c(ii)			(L8d)	(L11c)	
203 205a	mc23	L9c		L10c(ii) L10c(iii)				mc18							
204 205b					L7c			**L12d**		L6a(iii)					(L10c)
207															
208a															
208c	mc21 L8e	mc15	L10c(ii)	mc20	mc20		mc15		L11a(i) L11b(ii)B Part II	L7a(i)			(mc22)	L8b(iii)	mc16
209	mc21									mc17			L11a		
210			L7a(iii)				mc14	mc15	L11a(ii)	mc19 L8b(ii)					
212		mc17							L11a(iii)	L7b					
213					mc14	L8b(ii) L8b(iii) L8b(iv) L8b(v)		L12d(i) L12d(ii) L12d(iii)							
214													(mc25)		
215															
216		L9d		L10a L10b(ii)			mc17			L6a(iv)			(mc26)	L5a	
217															

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Section 2b: Synthesis Page 2

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
218 219 220	mc14						mc13	mc16	mc16					(mc20)	
221															
222															
223a															
223b 225	mc19 mc21	L11c				mc14	L10b(i) L10b(ii)	mc18					(L10e)		
223c 226a 227a		L9a(i)					L10c(i)		L4a(i)A L4a(i)B				(mc24)	(L8b(i))	
224a		L9a(ii)			mc13				L4b(i)	mc20					
208b 223d 226b 227b			L9b					L12b(i)						mc25	
224b								L12b(ii) L12b(iii)							
228					mc20										
229															
230															
231					mc15	L8c(ii)	mc16 L11c								
232															
233															
234		mc16	L7a(ii)	mc19		L8c(i)			mc17					mc24	
235								mc17						mc23	(mc17)
236 237	mc15				mc18					mc18					
238									L7b(ii)B						
239 240 241										mc16					
242	mc22 L9a(ii)	mc18	L10a	L8	L3a(i)										(L8a)
243					mc16 L3b(ii)	mc15 L11a(i)		L11a(i)		L11b(ii)B					
244			L7a(i)	mc21			L11a(ii) L11a(iii)							(L6c)	(L8b(ii))
245															
246	L8b				L3b(i) L3b(iii)	L8d(i)				L8b(i)			(L10a)	(L6a) (L6b)	(L8b(i))
247							L11a(i)	mc20	mc18				(mc27)		(L8c(ii))
248															

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Section 2c: Stereochemistry

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
249 250									**L4b(ii)**						
251 252															
253a 254a 255a			L4b L4c	mc17 mc18		L6b(i)	mc18 L9 L10b(iii)A L10b(iii)B	mc22 L6b(ii)	mc19					(mc16)	(L9c)
253b 254b 255b		L8b						mc21							
256						L6b(ii)									
257 259	L7b(i) L7b(ii)	L8c(i)	L7a(iv)	L9a(i) L9a(ii) L10d		mc19		mc22 L1b(i)	L1a	L10c(i)				(L9a)	(mc12)
258		L8c(ii)	mc17						L1b	mc22					
260 261 262					L6c				L1c						
263		mc13						L1b(ii)	L1d	L10c(ii)A				(mc17) (L11d)	(L10f)

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Section 2d: Experimental Determination of Structure

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
264 265															(L12a)
266 267	L10a		mc20	mc24	mc21 L6a	mc20	mc22	mc24	L5c(i) L5c(ii)	L12a			(L12a)		(L10a)
268															
269 270							mc19		L5a(ii)	L12b(i) L12b(ii)					
281 282	L10c(i) L10c(ii)	mc19	mc21	mc23	L6b(i) L6b(ii)	mc21							(L12b) (L12c)	(mc26) mc28	(L10b)
283 284				L7d(ii)A						L12c(i)				(mc27)	
285 286 287						L4b(i)				L12c(ii)					
288	L10b	mc20	mc22	L7d(ii)B	L12i	L4b(ii)	L11e(i)A L11e(i)B		L5b(i)				(L9b(i))	(mc29)	L11a(ii)
289						L4c(iii)		mc23							
291 292 293 294		L8a(ii) L11b(i)	L9c L10d		mc23	L4c(i) L4c(ii)	L11e(ii)A L11e(iii)A	L6b(i) L6b(iii)	L5b(ii)A L5b(ii)B				L7c		mc19 (L10e)
295								L6a(ii)							
296 297								L6a(i)	mc21						
298 299	mc24	L11b(ii)		mc25	mc17		L11e(ii)B	L6c		L12d			L12d		
300				L10e						mc23					
	L10d									L12c					L11c

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Section 2e: Pharmaceutical Chemistry

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
301 302 303					L9a										
304													(mc29)		
305 306 307	L8a	mc21	mc23	mc26	L9b	L10a	mc20	L2c(i)		L11b(i)			L7d	L7a	L11b
308							mc20								
309 310		mc21 L8d(i) L8d(ii)							L7b(ii)A	L11a(i) L11a(ii)					
311	mc25				L9d	L11a(ii)	mc21	mc25						(mc30)	

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Section 3a: Common Chemical Apparatus

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
312a															
312b															
312c															
312d															
312e															
312f															
312g															
312h															
312i							L6b(i)								
312j															
312k															
312l															
312m															
312n															

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Section 3b: Skills Involved in Practical Work

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
313a 313b 313c															
313d 313e															
313f															
313g		mc30													
313h													(L5b)	L2	
313i															
313j		mc26 L10a(iii)		L6d			L7d(ii)								L3c

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Section 3c: Stoichiometric Calculations

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
314 315		mc27 mc28				L8a	mc25								
316b 316c 316d 320d	mc28		mc24 L3b(i)						mc20	mc24			(mc12)		(L3b(i))
320a		mc29			L10a(i) L10a(ii) L10b	mc23									
320b 320c										mc25					
316a				L4c(ii)A	mc24		L8c		L2b(i)	L5b					
317						L3b(ii)		L2c(ii)							
318						L10b(i) L10b(ii) L10b(iii)			L10a(i)						
319	mc26	L4c	L7b	mc27	L9c	L2b	mc24	L2b L11b(ii)	L10b(iii)	L2b				L1e	
320e							L1b(ii)								
320f															
320h															
320g 321	L9b(v)	L7b		L9b(ii)	L7d(ii)	L8d(ii)	L6b(ii)			L8a(iii)			L11b(i)	(L7b(ii))	L4c
322													L11b(ii)	(L9b(ii))	

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Section 3d: Gravimetric Analysis

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
323 324	L4a(i) L4b(ii)	mc22	mc28 L6c(i)		L5b(i) L5b(ii)	mc23 L7a(i)		L3d(i) L3d(ii)	L3a						
325a 326	mc29	L6a(iii)A L6a(iii)B L6a(iii)C	L6c(ii)	mc3			L3a(i) L3a(ii) L3b L3c	L3d(iii)	L3b L3c	L3b(i) L3b(ii)			(mc11)	(L2b) L12	mc30
325b 327			L3b(ii)	mc2						L4❸					

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Section 3e: Volumetric Anaylsis

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
328	L4a(iii) L4c		L3a L3c		L4c										
329										L5a(ii)B					
330	L4a(ii)							L11b(i)					L9b		
331				L3a					L6a(i)				L9c		mc28
332						L7b(i)				L5a(ii)A			L9a		
333							mc23		mc22						
334 335	mc27													L2d	mc27
336														L2c	
337		L10a(i) L10a(ii)				L7b(ii)			L6a(ii)						
338	mc30			mc30	L5c(i)			L7b(i) L7b(ii)		L5a(i)A L5a(iii)			(L5b)		
339 340 341				L6a L6b	L4a L4b(i) L4b(ii)		L7d(i)								L3a (L3b)

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Past Paper Question Bank

Section 3f: Practical Skills & Techniques

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Outcome	AH 2016	AH 2017	AH 2018	AH 2019	AH 2020	AH 2022	AH 2023	AH 2024	AH 2025	AH 2026	AH 2027		revised AH 2013	revised AH 2014	revised AH 2015
342		mc23 L10a(iii)	L10c(i)A L10c(i)B	L3b(i) L3b(ii)	L5c(ii)			L9b(i)A					(L4a)		
343 344 345 346				L3b(iii)		L7a(ii)	L4	L9b(i)B L9b(ii)					(L4c)		
347 348 349			L8a L8b L8c		L12c	mc24									
350 351 352	L9b(i) L9b(ii)	L11a(ii)	mc30		L12c	L4a(i)A L4a(i)B				L8a(i)A L8a(i)B					
353 354 355	L4b(i)			L7a	L12f(ii)	L4a(iii)A	L6b(i)	L2a(ii)	L10b(ii)						
356 357 364	L9b(iv)	L7a(i)	mc26	mc28	L12g	L4a(iii)B	L11d	**L5**	mc25	L8a(ii)			mc21		L4b
358 359 362		mc25 L11a(i)		L7d(i)		L11b(i)A L11b(ii)		L2a(i)	L5d(i)				L5b		
360 361			mc25	mc29						L10b(ii)A					
363										L10b(ii)C					
365a 366 367															
368			mc27				L11e(iii)B			L8a(iv)					
365b 369		L7a(ii)			L12h(i) L12h(ii)				mc23						L4d
370 371 372			L7c			mc25									
373 374															
375 376		mc24	mc29												mc29
377															
378									L9d(iii)						L4e

Self-Evaluation Question Bank 2001-2015

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Past Paper Question Bank

Section 1a: Electromagnetic Radiation & Atomic Spectra

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
1 2 4											L2a				
3	mc1 mc22		mc1				mc3 mc5					mc1	mc2 mc3 mc4		mc1
5															
6 7 9															
8 13 14															
10 11 12		L1d		L5e	L13e	L4a		L3c	L1b	mc5 L1a	L2b	L1a(i)	L4d	mc1 mc2	L1b
15 16															
17 18					mc6										
19 20					mc4 mc5	mc1		L3a L3b	mc1			mc5			mc3
21												mc4	mc7	mc4	

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Past Paper Question Bank

Section 1b: Atomic Orbitals and Electronic Configurations

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
22 23 24															
25															
26						L3c	mc10								
27															
28a															
28b						L3d									
28c															
28d															
29a	mc11	g32b		mc1	mc3	mc7	mc4			mc1 mc3				mc3	
29b															L1a(ii)
29c			mc6				mc9						mc5		
30						L3b									L1a(i)
31															
32		g32a g32c				L3a			L1a						
33					mc1			mc1		mc2					
34 35 36				L1a L1b	mc2		L7b			mc4	mc1		mc1 mc6		
37 38						mc5	mc7			L1b(i) L5b	mc3		mc10		
39 40	L9a			mc6 mc7				L1a	mc4			L4a			
41 42	g32a g32b	mc21 L2a	L1c L3c	mc2	mc9	mc2	mc6		L12b	mc6 L1b(ii)	mc4 mc9	mc6	mc11	mc6 mc7 L1d	mc2 mc6
43	L9b	L2b						L1b							

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Past Paper Question Bank

Section 1c: Transition Metals

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
44															
45				L5a			L7a(i)								
46	L1a	L1c	mc7	mc8	L3b		L7a(ii)	mc2 mc11 L2b(iii)			L4a	mc2 mc3 L1a(ii)	L5b	L4e	mc2 L3b(iii)
47 48	mc5 L3b	mc19	mc4		L7a	L1b(i)	L9b(i)		L12a		L3a	L2a		mc8 L2c	L3b(i)
49 50															L3b(ii)
51 52				L5c				mc10					mc14		L3b(iv)
53							L9b(ii)								
54 55					L3a		L9a	L4a(i) L4a(ii)	L8c(i)			mc7			
57	mc9		mc5 L4a						L8c(ii)		L5b(ii)			L4a L9a	
56 58		mc18		L5b	L5a(i)		L9c	L2b(i) L2b(ii)	mc8	L2a L2b L2c	L4c	L2c	L5c	L4b	mc9
59 60 61	L1b L1c(i) L1c(ii)			L5d											
62															
63 64 65	mc2 mc3 L1c(iii)		mc2	mc9				L2a			mc10		L5a	L4c	
66 67 68															
69a				L7b(ii)					L7b						
69b 70															

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Past Paper Question Bank

Section 1d: Chemical Equilibrium

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
71 72	mc8	mc1 L4d	mc13	mc12 mc13	mc13	mc11		mc13		mc11 L7a	mc12		mc18 mc22	mc11	
73							L2a								
74 76		L4c	mc14				L2b		mc10	L7b	mc13	mc10		mc12 mc13	L5a(i)
75					mc21	mc21									
77															
78								mc12							
79					mc14		mc14	mc14	mc12			mc8			L5a(ii)
80															
81				mc2				mc3							
82			L3b(i)												
83														L1c	
84 85	L4a+b+c				mc15										
86												mc12			
87															
88			mc17					mc15							
89 90 91	L6a	L5b			L8a	L5c	L3a+b	mc16	L6a(i)	mc13	mc16	L7c			
92															
93 94															
95										L8a					
96															
97			mc16										mc21		mc14
98	mc24				L8b L8c	mc14 L5a		L9a L9b	L6a(ii)						
99		L10c	L11a			L5b	L3c		mc16 L6b	L8b		L7a	L6a		
100					mc16										
101															
102															
103				L8a							L9a				
104					mc17			L9c	mc15		L9b	mc11			mc13
105	L6b(iii)	L10d	L11b											L12b	
106	L6b(i) L6b(ii)			L8b		mc13					L9c		L6b	L12a	
107															
108			L12a												
109 110 111 112 113			L12b		mc18	mc15									
114	mc13	mc11	mc15					L9d		mc14		mc13	mc23		

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Section 1e: Reaction Feasibility

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
115															mc15
116															
117		L3a		L3a	mc19	L1a(i)	mc16	L5a(i)	L2b	L3a				L3a(i)	L2a
118 119 120 121	g31a L9d	mc13 mc16	mc26	mc21	mc20	mc20	mc20	mc19	mc18 mc19		mc23 mc24	mc21	mc27 mc28		mc21
122															
123 124															
125								mc18							mc18
126															
127		L3b	L10a			L1a(ii)	L1a	L5a(ii)	L2a	L3b			L2a	L3a(ii)	L2b
128	mc14				mc22 mc23			L5a(iii)		L3c					
129			mc22	mc22			mc22		mc17						mc19 mc20
130															
131	mc10			mc15									L2b		
132 135	g31b g31c	mc10		mc23	mc21										
133		L3c		L3b L3c		L1a(iii)	L1b		L2c			mc20		L3b	L2c
134															
136															

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Past Paper Question Bank

Section 1f: Kinetics

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
137															
138 139	L8c	L5a(i)	mc21		L7b(i)	L7b	mc24 L4a				L7b	L10a L10b	L8a	L10a L11d	mc26 L5b(i)
140 141 142		mc14 L5a(ii)	L9a		mc25 L7b(ii)	mc25	L4b		mc21				L8b	L10b(i)	
143				L7c(i)		L7c		L10d(i)	mc20	L9c	L7a				
144						L7a								mc24	mc25
145			L9b	L7c(ii)	L7b(iii)	L7d	L4c	L10d(ii)		L9d	L7c	L10c	L8c	L10b(ii)	L5b(ii)
146 147	mc7	L5a(iii)		mc27		mc24			L7a	L9a L9b				mc23	mc24
148	L8b							L10d(iii)						L10c	

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Past Paper Question Bank

Section 2a: Molecular Orbitals

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Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
149 150															
151 152															
153 154															
155 156 157															
158															
159							mc31							mc26	
160		L4b						L10b(ii)		mc28					
161									L12c						
162b 163					mc36	mc29					mc28				
162a 164 165								L10b(i)							
166												mc29			
167	L9c		mc32			mc28			mc26		mc30	mc30			
168 169															
170															
171 172															
173 174 175															
176															

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Past Paper Question Bank

Section 2b: Synthesis Page 1

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
177 178															
179 182a			mc29	mc29			mc25			mc26					
180 182b															
181 183															
184 185 186				mc31			mc29			mc10 mc31					
187 188 189									mc24						
190		mc28								mc30			mc30 mc39		
191															
192 193		L12d	mc36			mc35			mc25	mc29	L11b(i)	L12a			L9a L9c
194 196	L11a			mc32		mc26 L12c(i)	mc34	mc37		mc38 L13a		mc33 L4c	mc34		mc28 L10a+d
195 197															
198 199			mc35				mc33								
200 222				mc28				L10a			mc26		L10d	mc25	mc29 L10b
201a									L9b	L11c		mc23 L15a	L10c		
201b 211	mc23	mc27	L6c	L10b L10c(i)	L9a(i) L9b		mc36	mc31 L11a(i)+b	mc28	mc25 mc33	L13b(i)			mc29	mc32
201c		L10a+b		L9b	L10b	L12a+b		mc30		L12c				L8b(iii)	
202 206	L8a+d		L6a(i)+b	L10c(ii)	mc28	L12d(i)					L13a(i)	mc25	L8d	L10c	
203 205a									mc23		L13a(ii)				
204 205b			L6a(ii)		L11a	L12d(ii)					L13b(ii)	L15b			L10c
207															
208a															
208c		L12e	L5a(i)	L9c	mc26					g32d			mc29	L7b(ii) L7b(iii)	
209				L9a(i)		L8a	L12b								
210			mc34 L5b(ii) L5d	mc33	mc37 L9a(ii)	mc33 L9a L12c(ii)	mc35	L11a(ii) L12c(i)	mc31				L13b		
212											mc32				mc33
213						L10b(i) L10b(ii)		L11c				L11b			
214		mc26			mc29	L8b L10a		mc25 mc33		mc32	mc27		mc33		mc31
215															
216				L10a									mc35		
217															

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Past Paper Question Bank

Section 2b: Synthesis Page 2

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
218 219 220	mc6 L12c(i)		mc31			mc31		mc26 mc27 mc29	mc29					mc28	
221			L5b(i)												
222															
223a															
223b 225			mc39 L5a(ii)		mc35						mc31		L10e		
223c 226a 227a	L10c(i)	mc30	L5c			mc27	L12b(ii)	mc24	mc27	L11b		mc28	mc31	L7b(i)	mc30
224a	L10c(ii)	L12c													
208b 223d 226b 227b								L10c							
224b															
228															
229															
230				mc36			L10a L10b	mc32	mc30 mc32	L11e		L13b			
231															
232															
233															
234		mc5		L11c	L5b								L11b		
235			mc27					mc33	mc34	mc34			mc37		mc36
236 237	mc25		mc28				mc26 mc30	mc34			mc33				
238				mc37							mc34	mc34		mc34	
239 240 241															
242	L5a L5c			mc38		L11b	mc38	mc35	mc33	mc37		mc31			L8a
243															
244		L6a	mc33				mc27	L13a	mc35	mc36				L8c	L8b(ii)
245	mc15	L6b				L11a(i)		L13b			mc35				
246			mc38	L4c	L10a		L8c			L11d		mc35	L10a	L6a L6b	L8b(i)
247			mc37	mc30	mc40	L11a(ii)		L13c	L10a		mc36	L14a(i)	mc38		L8c(ii)
248						L11a(iii)		L13d				L14b(i)			

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Past Paper Question Bank

Section 2c: Stereochemistry

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
249 250															
251 252															
253a 254a 255a	L10a	L4a	L5e(i)	mc40	L11b	mc36		L4b	L9b	L11a	mc37		L13a	mc35	L9b
253b 254b 255b												mc36			
256															
257 259	L10b		L5e(ii) L11c(i)	L10d	L10c		L11a	mc36			L14b	L12b(i) L12b(ii)		L8a	mc37
258						L9b L9c			mc38	L12d(ii) L12d(iii)					
260 261 262															
263			L11c(ii)											mc36 L10d	L10f

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Section 2d: Experimental Determination of Structure

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
264 265			L11a L11b	L12b(i)			L13a(i)								L11a
266 267		mc4		L12b(ii)	L13b	mc37	L13a(ii) L13b	mc38	L13b(i)		L11a	mc37	L14a		L11a
268															
269 270		g31b								mc39					
281 282	L11b(i) L11b(ii)		L7b	L12c(i) L12c(ii)	L13c(i) L13c(ii)	mc38	mc40	L12c(iii)	L11a L11c		mc38		L14b L14c	mc37 mc38	L11b
283 284				mc39		mc39						mc39		mc39	
285 286 287												mc38			
288	mc27		L7a	L12a(i) L12a(ii)	L13a			mc39	L11b(i) L11b(ii)	mc40	L11b(ii)		L12c	mc33 L8b(i)	
289											mc39				
291 292 293 294			L7c			L13a	L13c(i)			L13b(i) L13b(ii)	mc40				L10e L11c
295						L13b	L13c(ii)								
296 297															
298 299													L14d		
300	L11c							L12c(ii)				L15c			
			L7d	L12d	L13d		L13c(i)		L11d						

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Past Paper Question Bank

Section 2e: Pharmaceutical Chemistry

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
301 302 303															
304													mc40		
305 306 307	mc30	L13b	L8a	L11a	L12a	mc40	L11c		mc40			mc40			mc39
308		L13a													
309 310			L8b					mc40							mc40
311			L8b	L11b	L12b		L11b				L10a L10b		L11c	mc40	

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Section 3a: Common Chemical Apparatus

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
312a															
312b															
312c															
312d															
312e															
312f															
312g															
312h															
312i															
312j															
312k															
312l															
312m															
312n															

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Section 3b: Skills Involved in Practical Work

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
313a 313b 313c															
313d 313e					L6a(i)										
313f															
313g												L5b			
313h	L2	L12b	L2	mc10 L6a L6b(i) L6b(ii)	L6	L2	L6	L6		L6	L4b L5b(iii)		L9b	L4f L5	mc10
313i															
313j															

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Past Paper Question Bank

Section 3c: Stoichiometric Calculations

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
314 315			mc10 mc11			mc10									
316b 316c 316d 320d	L2c(i)		mc9	mc11 L6b(i)	mc12 L6a(ii)			L6a	mc9	L6a(ii)	mc11		mc16 mc17		L7b(i)
320a															
320b 320c			L2a											L4f(i) L4f(ii)	
316a					L6a(iv)										
317	L2c(ii)			L6b(ii)											
318															
319															
320e							mc12								
320f		mc20													
320h															
320g 321							L12a(iv)			L10d		L13e		L9c	
322															

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Section 3d: Gravimetric Analysis

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
323 324					mc11				L8c(iv)						
325a 326		mc29				mc9	mc13		L8c(iii)				mc15	L5b	
325b 327															

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Section 3e: Volumetric Anaylsis

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
328	L2b					L2a		L8a	L7c(ii)	L6a(i)					
329															
330															
331															
332														mc10	
333															
334 335															
336					L6a(iii)										
337			mc12			L2c		L6b		L6a(iii)		L5a			
338						mc8	L6b				L5a L5b(iii)		L9b		
339 340 341			L2a L2b(i) L2b(ii)		L6b										L7b

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Section 3f: Practical Skills & Techniques

Copyright JAB chem

Outcome	Old AH 2001	Old AH 2002	Old AH 2003	Old AH 2004	Old AH 2005	Old AH 2006	Old AH 2007	Old AH 2008	Old AH 2009	Old AH 2010	Old AH 2011	Old AH 2012	Old AH 2013	Old AH 2014	Old AH 2015
342		L7		L6c					L8b		mc2		L4a		
343 344 345 346										L4b					
347 348 349															
350 351 352												L13a			
353 354 355															
356 357 364					mc30		L12a(iii)		mc13			mc27	L12b(i) L12b(ii)		
358 359 362						L8c	L12a(ii)								
360 361															
363															
365a 366 367															
368															
365b 369					L6c										
370 371 372											mc15				mc12
373 374															
375 376															
377															
378															